

C617

Chemistry: Content Knowledge

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A chemistry teacher who runs the school's lab-safety program asks how many grams of water form from 4.0 g of H_2 reacting with excess O_2 ($2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$). The reaction is demonstrated behind a blast shield. Use $\text{H}_2 = 2.0 \text{ g/mol}$ and $\text{H}_2\text{O} = 18.0 \text{ g/mol}$.

- A. 18 g
- B. 36 g
- C. 4.0 g
- D. 72 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An industrial lab with ISO certification mixes 2.0 mol N_2 with 3.0 mol H_2 for $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. The run was scheduled for the night shift. What is the maximum amount of NH_3 formed?

- A. 5.0 mol
- B. 2.0 mol
- C. 4.0 mol
- D. 6.0 mol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A lab technician at a hospital with 400 beds dilutes 50.0 mL of 6.0 M HCl to a final volume of 300.0 mL. The hospital recently switched suppliers. What is the new concentration?

- A. 36 M
- B. 0.17 M
- C. 1.0 M
- D. 6.0 M

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

How many mL of 12 M stock acid are needed to make 500 mL of 3.0 M acid?

- A.** 2,000 mL
- B.** 125 mL
- C.** 36 mL
- D.** 375 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A water-quality team at a state park with three lakes finds Lake A at pH 3 and Lake B at pH 5. The team's boat is newly painted. How does $[H^+]$ in Lake A compare with Lake B?

- A.** About 1.7 times greater
- B.** 1/100 as great
- C.** 100 times greater
- D.** 2 times greater

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A solution has $[H^+] = 1 \times 10^{-9}$ M at 25 °C. Which describes it?

- A.** Basic, pH 9
- B.** Neutral, because the concentration is tiny
- C.** Acidic, pH 9
- D.** Acidic, pH 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher at a school that just earned a STEM grant lists four compounds used in a lab kit: KBr, CH_4 , O_2 , and HCl. The kit arrived two weeks late. Which has ionic bonding?

- A.** HCl, since any polar bond is ionic
- B.** O_2
- C.** CH_4
- D.** KBr

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A student who tutors at the public library compares Li, Na, K, and Cl for a flashcard set of 60 cards. Which atom has the largest atomic radius?

- A. Li
- B. Cl
- C. Na
- D. K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which element has the highest first ionization energy: Li, Na, F, or Cs?

- A. Na
- B. Li
- C. F
- D. Cs

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A calorimetry lab at a university with a new chemistry building heats 200 g of water from 20 °C to 45 °C. The lab's coffee maker is next to the sink. With $c = 4.18 \text{ J/g}\cdot^\circ\text{C}$, how much heat did the water absorb?

- A. 20,900 J
- B. -20,900 J
- C. 836 J
- D. 37,620 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A reaction in a beaker has $\Delta H = -90 \text{ kJ}$. What does a student holding the beaker notice?

- A. It feels warmer; the reaction is exothermic.
- B. No change, because energy is conserved.
- C. It feels colder; the reaction is endothermic.
- D. It feels colder; the reaction is exothermic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A fertilizer plant with 700 workers runs $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at equilibrium, then raises the pressure. The plant's parking lot was recently expanded. Which way does the equilibrium shift?

- A.** Toward N_2 and H_2
- B.** No shift; pressure only speeds both reactions like a catalyst
- C.** No shift, because pressure never affects equilibrium
- D.** Toward NH_3 , the side with fewer gas moles

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A chemistry teacher ~~who runs the school's lab safety program~~ asks how many grams of water form from 4.0 g of H_2 reacting with excess O_2 ($2H_2 + O_2 \rightarrow 2H_2O$). ~~The reaction is demonstrated behind a blast shield.~~ Use $H_2 = 2.0$ g/mol and $H_2O = 18.0$ g/mol.

Target word: grams of water **Decides:** $2.0 \text{ mol } H_2 \rightarrow 2.0 \text{ mol } H_2O$ (2:2).

A	Dropped element	Drops the moles and uses one mole of water.
B	Correct	$4.0 \div 2.0 = 2.0 \text{ mol } H_2$; 2:2 ratio gives $2.0 \text{ mol } H_2O \times 18.0 = 36 \text{ g}$.
C	Wrong stem number	Carries the reactant mass straight across.
D	Inverted operation	Doubles the moles by misreading the 2:2 ratio as 1:2.

2. Answer B Concept 2

An industrial lab ~~with ISO certification~~ mixes 2.0 mol N_2 with 3.0 mol H_2 for $N_2 + 3H_2 \rightarrow 2NH_3$. ~~The run was scheduled for the night shift.~~ What is the maximum amount of NH_3 formed?

Target word: maximum product **Decides:** H_2 needs 6.0 mol but has 3.0: H_2 limits.

A	Wrong stem number	Adds the reactant moles.
B	Correct	$3.0 \text{ mol } H_2 \times (2 \text{ } NH_3 / 3 \text{ } H_2) = 2.0 \text{ mol}$.
C	Dropped element	Uses N_2 as limiting because it is the smaller number.
D	Inverted operation	Multiplies H_2 moles by 2 without dividing by 3.

3. Answer C Concept 3

A lab technician ~~at a hospital with 400 beds~~ dilutes 50.0 mL of 6.0 M HCl to a final volume of 300.0 mL. ~~The hospital recently switched suppliers.~~ What is the new concentration?

Target word: diluted molarity **Decides:** $M_2 = M_1V_1/V_2$.

A	Inverted operation	Multiplies by the volume ratio upside down: $6.0 \times 300 \div 50$.
B	Dropped element	Drops the 6.0 M: $50 \div 300$.
C	Correct	$6.0 \times 50.0 \div 300.0 = 1.0$.
D	Absolute language	Adding water always lowers concentration.

4. Answer B Concept 3 • transfer

How many mL of 12 M stock acid are needed to make 500 mL of 3.0 M acid?

Target word: solve for stock volume **Decides:** $V_1 = M_2V_2/M_1$.

A	Inverted operation	Inverts the ratio: $12 \times 500 \div 3$.
B	Correct	$3.0 \times 500 \div 12 = 125$.
C	Wrong stem number	Multiplies the two molarities.
D	Confusion set	That is the water added, not the stock needed.

5. Answer C Concept 4

A water-quality team at a state park with three lakes finds Lake A at pH 3 and Lake B at pH 5. The team's boat is newly painted. How does $[H^+]$ in Lake A compare with Lake B?

Target word: $[H^+]$ ratio **Decides:** Two pH units = 10^2 .

A	Wrong stem number	Divides 5 by 3.
B	Inverted operation	Lower pH means more H^+ , not less.
C	Correct	$10^{5-3} = 10^2 = 100$.
D	Confusion set	Treats the pH difference as linear.

6. Answer A Concept 4 • transfer

A solution has $[H^+] = 1 \times 10^{-9}$ M at 25 °C. Which describes it?

Target word: classify from $[H^+]$ **Decides:** pH = 9.

A	Correct	pH = $-\log(10^{-9}) = 9 > 7$.
B	Absolute language	Neutral is pH 7, $[H^+] = 10^{-7}$ M.
C	Inverted operation	pH above 7 is basic.
D	Confusion set	5 is the pOH, and it is still basic.

7. Answer D Concept 5

A teacher at a school that just earned a STEM grant lists four compounds used in a lab kit: KBr, CH_4 , O_2 , and HCl. The kit arrived two weeks late. Which has ionic bonding?

Target word: ionic compound **Decides:** Metal plus nonmetal.

A	Absolute language	HCl is polar covalent; polarity is not ionic transfer.
B	Adjacent category	Identical atoms share equally: nonpolar covalent.
C	Confusion set	Nonmetals sharing electrons: covalent.
D	Correct	Potassium (metal) transfers an electron to bromine (nonmetal).

8. Answer D Concept 6

A student who tutors at the public library compares Li, Na, K, and Cl for a flashcard set of 60 cards. Which atom has the largest atomic radius?

Target word: largest radius **Decides:** Radius grows down a group; K is lowest in Group 1.

A	Confusion set	Li is at the top of the group, the smallest of the Group 1 atoms.
B	Inverted operation	Radius decreases across a period; Cl is small.
C	Dropped element	Ignores that K sits one period lower.
D	Correct	Potassium has the most occupied energy levels of the group listed.

9. Answer C Concept 6 • transfer

Which element has the highest first ionization energy: Li, Na, F, or Cs?

Target word: highest ionization energy **Decides:** Increases up and to the right.

A	Dropped element	Mid-group; lower than F.
B	Confusion set	Top of Group 1 but far left of F.
C	Correct	Top right of the set, holding electrons most tightly.
D	Inverted operation	Largest atom, lowest ionization energy.

10. Answer A Concept 7

A calorimetry lab at a university with a new chemistry building heats 200 g of water from 20 °C to 45 °C. The lab's coffee maker is next to the sink. With $c = 4.18 \text{ J/g} \cdot ^\circ\text{C}$, how much heat did the water absorb?

Target word: heat absorbed **Decides:** $\Delta T = 25 \text{ }^\circ\text{C}$.

A	Correct	$200 \times 4.18 \times 25 = 20,900$.
B	Inverted operation	The water absorbed heat, so q is positive.
C	Dropped element	Drops ΔT .
D	Wrong stem number	Uses the final temperature 45 as ΔT .

11. Answer A Concept 7 • transfer

A reaction in a beaker has $\Delta H = -90 \text{ kJ}$. What does a student holding the beaker notice?

Target word: sign of ΔH to observation **Decides:** Negative ΔH releases heat.

A	Correct	Energy flows out to the surroundings.
B	Absolute language	Energy is conserved but transferred to the surroundings.
C	Inverted operation	Reverses the sign meaning.
D	Confusion set	Exothermic warms the surroundings.

12. Answer D Concept 8

A fertilizer plant with 700 workers runs $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ at equilibrium, then raises the pressure. The plant's parking lot was recently expanded. Which way does the equilibrium shift?

Target word: shift direction under pressure **Decides:** 4 gas moles left, 2 right.

A	Inverted operation	Reverses the fewer-moles rule.
B	Confusion set	Confuses a stress with a catalyst, which does not shift equilibrium.
C	Absolute language	Pressure matters when gas moles differ.
D	Correct	Higher pressure favors fewer gas moles.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	5	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Stoichiometry through moles	Convert grams to moles, use the balanced equation's mole ratio, then convert back to grams.	Using grams directly in the mole ratio or ignoring coefficients.	"How many grams of product from ... grams of reactant?"
2	Limiting reactant	The limiting reactant is the one that runs out first based on the mole ratio, not the one with fewer moles or grams.	Picking the reactant present in the smaller amount.	Amounts given for two reactants.
3	Dilution	Moles of solute stay constant on dilution: $M_1V_1 = M_2V_2$.	Thinking dilution does not change concentration, or inverting the ratio.	A stock solution and a final volume.
4	pH is logarithmic	$\text{pH} = -\log[\text{H}^+]$; each unit is a $10\times$ change in $[\text{H}^+]$, and lower pH means more acidic.	Treating pH differences as linear.	Two pH values compared.
5	Bond type	Ionic bonds form between metals and nonmetals by electron transfer; covalent bonds form between nonmetals by sharing, polar when electronegativities differ.	Calling any polar bond ionic, or any compound with hydrogen ionic.	A list of compounds and "which is ionic?"
6	Periodic trends	Atomic radius increases down a group and decreases across a period; ionization energy and electronegativity follow the opposite trend.	Reversing the across-a-period trend because atomic number rises.	"Largest atom" or "highest ionization energy".
7	Thermochemistry	$q = mc\Delta T$ with $\Delta T = \text{final} - \text{initial}$; a negative ΔH is exothermic (surroundings warm), positive is endothermic.	Using the final temperature for ΔT , or mixing up the sign of ΔH .	Mass, specific heat, and two temperatures, or a ΔH sign.
8	Le Chatelier's principle	A system at equilibrium shifts to partly offset a stress; higher pressure favors the side with fewer gas moles, and a catalyst does not shift equilibrium.	Thinking pressure never matters, or that a catalyst shifts the position.	A stress applied to an equilibrium with gas moles on each side.

Course scope (public description): Chemistry: Content Knowledge provides advanced instruction in the main areas of chemistry for which secondary chemistry teachers are expected to demonstrate competency. Topics include matter and energy, thermochemistry, structure, bonding, reactivity, biochemistry and organic chemistry, solutions, the nature of science and technology, mathematics, and laboratory procedures.